Hybrid Retrieval and Neural Reranking for Portuguese Regulatory Search: A Zero-Shot Evaluation on ANEEL Resolutions

Abstract

Regulatory institutions increasingly rely on retrieval systems to locate norms, obligations, and procedural rules in large document collections, yet there is little empirical evidence on how modern retrieval pipelines behave on Brazilian Portuguese regulatory text. We evaluate zero-shot retrieval and neural reranking pipelines for Portuguese regulatory search using resolutions from the Brazilian Electricity Regulatory Agency (ANEEL) at two granularities: full documents and contextualized chunks. We compare lexical retrieval (BM25), dense retrieval (Qwen3-Embedding), hybrid sparse-dense retrieval with Reciprocal Rank Fusion, and off-the-shelf multilingual cross-encoder rerankers, in single-stage and multi-stage configurations, over an LLM-generated known-item benchmark with 1,000 synthetic queries. Lexical evidence proves essential: BM25 outperforms dense retrieval at the document level, while hybrid retrieval yields the strongest first-stage candidates at both granularities. Combined with a single BGE reranker, the hybrid pipeline achieves the best MRR@10 (0.794 for documents, 0.566 for chunks) and the best chunk-level Recall@10 (0.790), while hybrid retrieval alone retains the best document-level Recall@10 (0.934 vs. 0.926 after reranking); deeper Qwen-based reranking cascades increase latency without proportional gains. These results indicate that hybrid lexical-dense retrieval followed by lightweight reranking offers a more effective and practical trade-off than deeper neural cascades for Portuguese regulatory search.

Keywords: Regulatory information retrieval, Hybrid retrieval, Neural reranking, Portuguese, Zero-shot evaluation, Synthetic queries

1. Introduction

Legal and regulatory institutions increasingly depend on large-scale digital repositories to support decision-making, auditing, compliance analysis,

and legal research. In these settings, retrieval systems must identify relevant documents from heterogeneous collections containing decisions, reports, legal opinions, technical statements, regulatory acts, and procedural documents. Unlike general-purpose web search, legal and regulatory search requires high precision at the top of the ranking, since relevance may depend on legal authority, procedural context, temporal validity, institutional terminology, and the relationship between facts, rules, and precedents [1, 2].

Retrieval in this domain must reconcile two partially conflicting requirements. On the one hand, regulatory relevance often hinges on exact matches: resolution numbers, article references, institutional identifiers, deadlines, and fixed technical expressions. Lexical methods such as BM25 [3] directly exploit these signals. On the other hand, users express information needs in natural language that rarely mirrors formal legal-administrative wording, so relevance also depends on semantic matching across different formulations of the same regulatory issue. Dense retrieval models [4] address this second requirement by mapping queries and documents to vector representations, but may struggle with exact regulatory references and fine-grained top-rank ordering [5], especially when documents are long and partially overlapping.

Hybrid retrieval combines sparse lexical evidence and dense semantic similarity, typically through rank fusion techniques such as Reciprocal Rank Fusion [6, 7]. This design is attractive for regulatory search because lexical retrieval captures formal references and domain-specific terminology, while dense retrieval recovers semantically related candidates that do not share the same surface form as the query. Despite the growing use of dense and hybrid retrieval in retrieval-augmented generation and institutional search systems, existing Portuguese retrieval benchmarks [8, 9, 10] do not cover regulatory resolutions, and there is still limited empirical evidence on how these alternatives behave in Brazilian Portuguese regulatory corpora, especially when combined with neural reranking.

Neural rerankers based on cross-encoders [11] can refine candidate rankings by jointly processing each query-candidate pair and assigning a relevance score from token-level interactions. This expressiveness comes with substantial computational cost, since every candidate must be processed at inference time. Multi-stage pipelines address this cost by progressively reducing candidate sets before applying more expensive models [12, 13, 14, 15]. Yet additional stages may introduce early pruning errors that later stages cannot recover, and larger models may increase latency without proportional gains. These trade-offs are especially relevant for long Portuguese regula-

tory documents, where evidence may be distributed across articles, annexes, definitions, exceptions, and procedural conditions.

In this paper, we evaluate zero-shot retrieval and reranking pipelines for Portuguese regulatory information retrieval in the Brazilian electric power sector, using resolutions from the Brazilian Electricity Regulatory Agency (ANEEL) — a domain that combines legal, administrative, technical, and economic language. We deliberately do not fine-tune the evaluated models: our goal is to understand how publicly available retrievers and rerankers behave when deployed directly in a specialized Portuguese domain, a deliberately bounded deployment scenario in which no annotated data or domain-specific ranking models are assumed to be available. The evaluation is organized around three research questions:

RQ1 How do zero-shot lexical and dense retrievers compare on Portuguese regulatory search, and does the comparison change between document-level and chunk-level granularity?

RQ2 Does hybrid sparse-dense retrieval improve first-stage candidate generation over its individual components across both granularities?

RQ3 How much does neural reranking add over each first-stage strategy, and do deeper multi-stage reranking cascades justify their additional computational cost?

The contributions of this work map directly onto these questions. First, we present an empirical zero-shot comparison of lexical, dense, hybrid, and neural reranking pipelines for Brazilian electric-sector regulatory retrieval at two granularities (RQ1–RQ3). Second, we show that lexical retrieval is not merely a baseline but a competitive component of the retrieval architecture, and that hybrid sparse-dense fusion produces the strongest first-stage candidate sets (RQ1, RQ2). Third, we analyze the quality-latency trade-off of neural reranking over dense, lexical, and hybrid candidates, showing that a simple retrieve-then-rerank architecture can outperform deeper cascades in the evaluated setting (RQ3). Fourth, we document an LLM-generated known-item benchmark for ANEEL regulatory search, including the generation protocol, a detailed characterization of the generation prompt (Appendix Appendix A), and human validation of a sample of synthetic queries (Section 3).

Rather than proposing a new ranking model, this work provides evidence on how off-the-shelf retrieval and reranking components behave in a Portuguese regulatory search setting built from real ANEEL resolutions, and derives evidence-based guidance for institutional deployments over long and formal documents, where architectural choices must be justified against both effectiveness and latency constraints.

The remainder of this paper is organized as follows. Section 2 positions our study within legal and Portuguese information retrieval. Section 3 describes the ANEEL corpus and the construction and validation of the benchmark. Section 4 details the evaluated pipelines and the experimental protocol. Section 5 reports results by research question, Section 6 discusses their implications and limitations, and Section 7 concludes.

2. Related Work

2.1. Legal and regulatory information retrieval

Legal Information Retrieval differs from general-purpose search because relevance is not determined only by topical similarity or lexical overlap. Legal documents are often long, technical, and structured around procedural, normative, and argumentative elements, so relevance may depend on legal authority, jurisdiction, procedural stage, temporal validity, institutional terminology, and the relationship between facts, rules, and precedents. van Opijnen and Santos [1] argue that legal relevance has several distinct dimensions, while Sansone and Sperlí [2] highlight open challenges related to semantic understanding, scalability, and the integration of modern NLP methods into legal search workflows. Within the regulatory subdomain specifically, Rayo Mosquera et al. [16] combine lexical and dense retrieval with answer generation over regulatory texts, reporting effective hybrid retrieval in their evaluated regulatory collection. These studies establish the setting we target, but, to our knowledge, none of them evaluates the interaction between first-stage retrieval strategy, reranking depth, and latency in Brazilian Portuguese regulatory corpora — the combination our study addresses.

Pre-trained language models have substantially influenced legal NLP. Domain-specific models such as LEGAL-BERT show that pre-training on legal corpora improves legal language understanding [17], and LexGLUE systematizes evaluation across legal NLP tasks [18]. In the Brazilian context, Juru adapts large language models to Brazilian legal Portuguese [19]. This line of work demonstrates the value of domain adaptation for legal *language*

understanding, but it leaves open the question we pose: how well *off-the-shelf* retrieval components perform zero-shot in this setting, without any domain adaptation.

2.2. First-stage retrieval and neural reranking

Modern semantic search commonly separates retrieval into at least two stages: a first-stage retriever — sparse [3], dense bi-encoder [4], or a hybrid fusion of both [6, 7] — followed by one or more reranking stages [5]. Cross-encoder rerankers [11, 20] are effective because they jointly encode the query and each candidate, but each query-candidate pair must be processed at inference time. The resulting effectiveness-efficiency trade-off has long been recognized in learning-to-rank [12] and in neural ranking more broadly [21]. Multi-stage ranking mitigates inference cost by progressively pruning candidates [13], and later work explores multi-stage reranking variants and full-pipeline optimization [14, 15]. In legal retrieval, retrieve-then-rerank designs appear in COLIEE systems [22, 23] and BERT-based two-stage ranking [24]. For Portuguese legal documents, Novais et al. [25] evaluate cascade reranking on a large legal corpus from a Brazilian court of accounts, optimizing latency at a small effectiveness cost. In contrast to these works, which tune or optimize a chosen cascade, we ask a prior question: whether deeper off-the-shelf cascades are worth their cost at all in a zero-shot regulatory setting, compared against strong lexical and hybrid baselines evaluated under identical conditions.

2.3. Portuguese IR benchmarks and synthetic evaluation

Reproducible evaluation for Portuguese retrieval has advanced quickly. mMARCO extends MS MARCO passage ranking to Portuguese via machine translation [26], and Quati provides queries written by native Brazilian Portuguese speakers [8]. In the legal domain, JurisTCU contributes query relevance judgments over documents of the Brazilian Federal Court of Accounts [9], and JUÁ proposes a benchmark across Brazilian legal text collections [10]. To our knowledge, none of these resources covers regulatory resolutions of the electric power sector, and none evaluates the retrieval-plus-reranking pipeline space we study.

Because manual relevance annotation is expensive, synthetic query generation and LLM-assisted evaluation are increasingly used in retrieval research. InPars [27] and Promptagator [28] generate synthetic queries with LLMs primarily to train retrievers — a setting distinct from, though methodologically

related to, benchmark construction; Thomas et al. [29] show that LLM relevance judgments can approximate human preferences at scale; Faggioli et al. [30] discuss the methodological limits of LLM-based judgment; and UMBRELA operationalizes LLM assessment in large evaluation campaigns [31]. Zero-shot evaluation across heterogeneous domains is standardized by BEIR [32]. Our benchmark follows this synthetic-generation methodology and inherits its known limitations, which we state explicitly in Section 3 and examine in Section 6.5.

2.4. Positioning

In summary, prior work establishes (i) the specificity of legal and regulatory relevance, (ii) the effectiveness-efficiency tension of neural reranking cascades, and (iii) synthetic query generation as a practical route to benchmark construction, with known limitations. To our knowledge, no prior study combines these threads for Brazilian Portuguese regulatory search: a controlled zero-shot comparison of lexical, dense, hybrid, and single- versus multi-stage reranking pipelines, at two retrieval granularities, with jointly reported effectiveness and latency, over a documented synthetic benchmark with partial human validation. This combination is the gap our study fills.

3. The ANEEL Regulatory Benchmark

3.1. Corpus

We conduct our experiments on a corpus crawled from the public document library of ANEEL, the Brazilian Electricity Regulatory Agency. The exact collection period and the crawling and text-extraction procedure of the snapshot are not documented in this study; we register this as a reproducibility limitation in Section 6.5. The corpus contains four families of resolutions: Authorizing Resolutions (REA), Homologatory Resolutions (REH), Normative Resolutions (REN), and generic Resolutions (RES). These documents are long and heterogeneous, combining legal, regulatory, administrative, technical, and economic language.

We build two parallel views of the corpus. The chunk-level view segments the raw documents into contextualized chunks, resulting in 134,520 unique chunks across the four document families. The document-level view concatenates chunks from the same source document, resulting in 18,364 complete documents, each corresponding to the full text of an ANEEL resolution. Chunks target approximately 1,000 words with an overlap of 150 words and

respect the structural boundaries of the resolutions in priority order: articles of at least 120 words and tables are kept intact, followed by headings, paragraphs, and sentences. Chunks shorter than 120 words are merged with adjacent content, and revoked blocks are kept separate from the text in force.

We use contextual chunking to reduce the loss of meaning that occurs when a passage is separated from its source document, following the contextual retrieval strategy proposed by Anthropic [33]. Each chunk receives a short LLM-generated prefix situating it within the source document — the document topic, the surrounding section, or the regulatory matter under discussion — and the text indexed by retrievers and processed by rerankers is the contextualized chunk, not the raw chunk. This design is especially important in regulatory documents, where isolated paragraphs may depend on article numbers, dates, legal references, previous definitions, or surrounding procedural context.

For the benchmark, we sample 500 chunks and 500 documents from the full corpus, using stratified sampling by document type and token-length bin so that the benchmark reflects the distribution of the original corpus and avoids concentrating examples in a single family or length range.

3.2. Synthetic query generation

We formulate the benchmark as a question-based retrieval task. For each sampled item, we generate one synthetic question that serves as the retrieval query; the sampled item is the expected target. For chunk retrieval the target is the sampled chunk, and for document retrieval it is the source ANEEL resolution.

Synthetic query generation with LLMs has been used to construct retrieval training and evaluation sets when manual annotation is unavailable [27, 28], and LLM-based relevance assessment has been shown to approximate human judgments at scale [29, 31]. It also carries known limitations — generated queries may not reflect real user behavior and may share surface form with the source passage [30] — which our prompt design attempts to mitigate and our validation study quantifies.

Questions were generated with Gemini 3 Flash [34] through the `google-genai` SDK in asynchronous mode: one question per sampled passage, a 120-second timeout per call, up to six retries with exponential backoff, and maximum concurrency of 100 calls. Generation temperature followed the API default.

The prompt, characterized in detail in Appendix Appendix A, instructs the model to produce exactly one self-contained question that sounds like a real user query, can be answered from the provided text, and avoids meta-references such as "text", "excerpt", "document", "above", or "below". It also discourages copying phrases from the source, using internal identifiers, or producing questions that only make sense to someone reading the passage. The verbatim prompt string is part of the generation code and is covered by the data availability statement. We generate two question types in balanced proportion: *general* questions about the broader topic, objective, or regulatory matter of the target, and *specific* questions about concrete details such as deadlines, conditions, obligations, responsible entities, consequences, or exceptions.

Each benchmark instance follows the same schema: a `base_id`, the indexed `text`, and the `label` used as the query. During evaluation, we submit the question to each pipeline and check whether the expected `base_id` appears among the top-ranked results.

3.3. Human validation

To assess the quality of the synthetic benchmark, we performed a human validation study on a stratified sample of 50 of the 1,000 generated questions (5%), with 25 document-level and 25 chunk-level questions. A single annotator — one of the authors — performed the classification; the implications of single-annotator judgment are discussed in Section 6.5. Each question was classified as *valid* (natural, self-contained, answerable from the target passage, not a near-literal copy), *partial* (usable but with a minor issue: somewhat ambiguous, overly generic, or dependent on implicit context), or *bad* (not answerable from the target, copied too closely, referred explicitly to the text, or too generic to identify a meaningful target).

Overall, 86% of the sampled questions were judged valid, 8% partial, and 6% bad. All document-level questions were judged valid; at the chunk level, 72% were valid, 16% partial, and 12% bad — that is, 28% of the inspected chunk-level questions had some defect. In the inspected sample, the imperfect chunk-level cases were mainly associated with the contextualization process: some questions were answerable from the contextual prefix or the broader document, but not fully from the specific chunk. This observation matters when interpreting chunk-level results, since some retrieval failures may reflect benchmark granularity rather than model quality. The validation covers

5% of the queries; the implications of this partial coverage are examined in Section 6.5.

3.4. Task formulation

The benchmark is a *known-item* retrieval task: each query has exactly one relevant target — the item from which it was generated. This formulation has two direct consequences for metric interpretation. First, Recall@10 is equivalent to a top-10 success rate (the fraction of queries whose single target appears in the top 10). Second, the evaluation does not credit the retrieval of other relevant items, which are likely to exist in a regulatory corpus where multiple resolutions may address the same matter. We adopt this formulation deliberately — it requires no manual relevance annotation and yields a fully reproducible target set, in line with synthetic-benchmark practice [32] — and we discuss its limitations, including paths toward multi-target relevance judgments, in Section 6.5.

4. Experimental Setup

4.1. First-stage retrieval

We evaluate three first-stage strategies, each applied separately to the document-level and chunk-level corpus views.

Lexical retrieval.. The lexical baseline uses BM25 [3], implemented with the `rank_bm25` library (`BM25Okapi`) under its default parameters ($k_1 = 1.5$, $b = 0.75$), a lowercase `\w+` tokenizer, a compact list of approximately 60 Portuguese stopwords, and no stemming or accent removal. This baseline is particularly meaningful in regulatory search, where exact terms, institutional identifiers, resolution numbers, article references, dates, tariff classes, and domain-specific expressions are strong relevance signals; it also provides an interpretable reference point for assessing whether dense retrieval actually improves candidate generation in this domain.

Dense retrieval.. The dense retriever uses `Qwen/Qwen3-Embedding-0.6B` [35], which produces 1024-dimensional vectors and supports inputs up to 8,192 tokens. We use the recommended query prompt for queries and no prompt for documents. All vectors are L2-normalized and indexed with a FAISS `IndexFlatIP` index [36], so that inner-product search is equivalent to cosine similarity; separate indices are built for chunks and documents.

Hybrid retrieval.. The hybrid retriever combines BM25 and dense retrieval using Reciprocal Rank Fusion [6]. For each query we retrieve the top-100 candidates from each retriever and fuse the rankings as

$$\mathrm{RRF}(d) = \sum_{r \in R} \frac{1}{k + \mathrm{rank}_r(d)},$$

where R is the set of rankings, $\mathrm{rank}_r(d)$ is the position of candidate d in ranking r, and $k = 60$ as in Cormack et al. [6]. Candidates absent from one ranking contribute only through the ranking in which they appear.

4.2. Cross-encoder rerankers

After first-stage retrieval, we evaluate cross-encoder rerankers [11]. Given a query q and candidate d_i, the reranker jointly encodes the pair and assigns a score $s(q, d_i)$ from token-level interactions; candidates are sorted by this score. Unlike bi-encoders, cross-encoders cannot precompute document representations, so every query-candidate pair is processed at inference time.

We use three off-the-shelf multilingual rerankers of different sizes and context limits: `BAAI/bge-reranker-v2-m3` [37], with maximum input length 1,024 tokens; `Qwen/Qwen3-Reranker-0.6B`; and `Qwen/Qwen3-Reranker-4B` [35], both with maximum input length 4,096 tokens. Rerankers are loaded as `sentence_transformers.CrossEncoder` models and executed on GPU. Maximum input length is an important experimental factor: ANEEL resolutions often exceed the rerankers' context windows, especially at the document level, in which case the model scores a truncated query-candidate pair. This makes the evaluation directly relevant for long-document regulatory retrieval, where reranking may fail to help if the decisive evidence falls outside the observed context.

4.3. Evaluated architectures

We compare retrieval-only baselines, single-stage BGE reranking, and Qwen-based multi-stage pipelines (Figure 1). The retrieval-only baselines return the top-10 results from BM25, dense, or hybrid retrieval. The single-stage configurations retrieve the top-100 candidates from one first-stage retriever and apply `bge-reranker-v2-m3` to produce the final top-10. The multi-stage pipelines use dense retrieval as first stage: in the two-stage pipeline, `Qwen3-Reranker-0.6B` reduces the top-100 candidates to 30,

and `Qwen3-Reranker-4B` produces the final top-10; in the three-stage cascade, BGE keeps the top-40, `Qwen3-Reranker-0.6B` keeps the top-15, and `Qwen3-Reranker-4B` returns the final top-10.

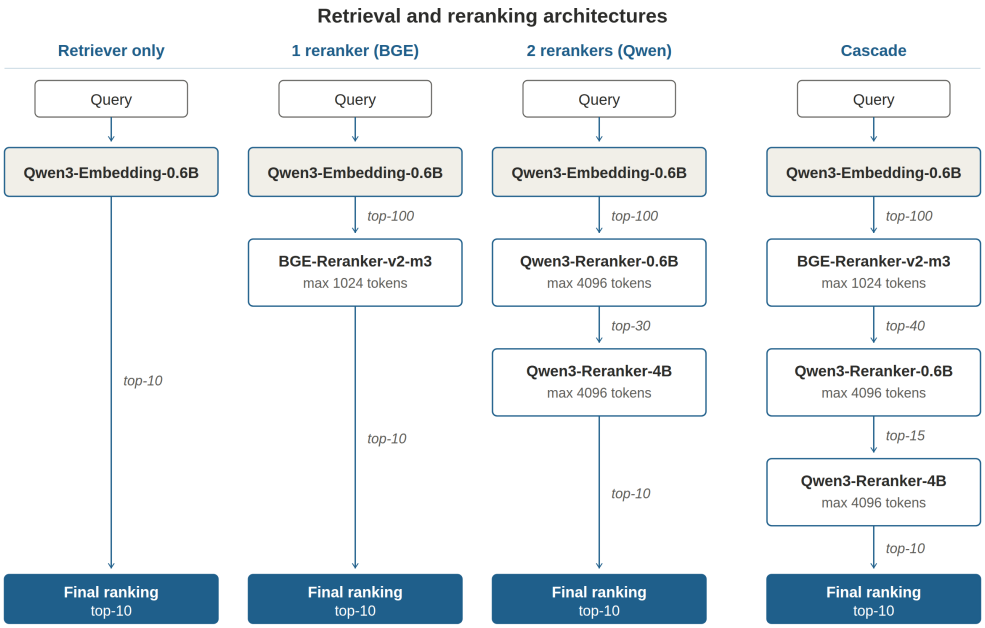


Figure 1: Retrieval and reranking architectures evaluated in this work: lexical, dense, and hybrid first-stage retrieval; single-stage BGE reranking over each first-stage retriever; and Qwen-based multi-stage pipelines using dense retrieval as the first stage. Arrows indicate candidate flow; numbers indicate the cutoff applied after each stage.

Formally, a reranking stage is

$$C_j = \mathrm{Top}_{N_j}\left(\{(d,\, s_j(q,d)) \mid d \in C_{j-1}\}\right),$$

where C_{j-1} is the candidate set received by stage j, s_j is the stage's scoring function, and N_j is the cutoff applied after scoring. This formulation makes each stage's candidate budget explicit and supports comparison across the predefined budgets. Throughout the paper, stage counts refer to reranking stages applied after first-stage retrieval.

4.4. Metrics and measurement protocol

We evaluate ranking quality with MRR@10 and Recall@10. MRR@10 averages reciprocal ranks over queries, assigning zero when the target is absent

from the top 10; Recall@10 is the fraction of queries whose target appears in the top 10 — under our known-item formulation (Section 3.4), a top-10 success rate. We report both because they capture complementary aspects: whether the system finds the target among results a user is likely to inspect, and how early it places the target in the ranking.

Latency is measured as average wall-clock time per query, including all online stages required to produce the final top-10. Dense retrieval runs on FAISS, lexical retrieval on BM25, and cross-encoder inference on an NVIDIA H100 80GB GPU, with `torch.cuda.synchronize()` around each reranking stage for faithful GPU timing. Only average latency per query is reported; run-to-run variability was not assessed. All architectures are evaluated with the same query sets, target identifiers, corpus views, and metric implementation, separately for each granularity. An architecture is considered preferable when it improves MRR@10 or Recall@10 without disproportionate latency; an architecture that increases inference time without improving quality over a simpler alternative is considered dominated in the evaluated setting.

5. Results

Table 1 presents effectiveness results for all evaluated configurations at both granularities; we organize the analysis by research question.

5.1. RQ1: Lexical versus dense retrieval

Lexical matching is a strong signal in Brazilian regulatory search, and its advantage depends on granularity. At the document level, BM25 outperforms the dense Qwen retriever, increasing MRR@10 from 0.656 to 0.677 and Recall@10 from 0.860 to 0.914. This pattern is consistent with the importance of exact terms, institutional identifiers, dates, tariff references, and regulatory expressions in ANEEL resolutions, although the aggregate metrics do not isolate which of these signals drives the advantage. At the chunk level the picture inverts slightly: dense retrieval reaches 0.364 MRR@10 against 0.347 for BM25, with nearly identical Recall@10 (0.610 vs. 0.612). Lexical and semantic retrieval therefore behave differently across granularities, and neither dominates the other on its own.

5.2. RQ2: Hybrid first-stage retrieval

Hybrid retrieval provides the strongest first-stage candidate generation at both granularities. At the document level, the hybrid Qwen+BM25 retriever

Granularity	Architecture	MRR@10	R@10
Docs	Dense (Qwen)	0.656	0.860
Docs	BM25	0.677	0.914
Docs	Hybrid (Qwen+BM25)	0.755	**0.934**
Docs	Dense (Qwen) + BGE	0.792	0.922
Docs	BM25 + BGE	0.787	0.912
Docs	Hybrid (Qwen+BM25) + BGE	**0.794**	0.926
Docs	Dense + 2RR (Q0.6B→Q4B)	0.644	0.708
Docs	Dense + Cascade (BGE→Q0.6B→Q4B)	0.649	0.712
Chunks	Dense (Qwen)	0.364	0.610
Chunks	BM25	0.347	0.612
Chunks	Hybrid (Qwen+BM25)	0.432	0.698
Chunks	Dense (Qwen) + BGE	0.554	0.760
Chunks	BM25 + BGE	0.555	0.748
Chunks	Hybrid (Qwen+BM25) + BGE	**0.566**	**0.790**
Chunks	Dense + 2RR (Q0.6B→Q4B)	0.555	0.736
Chunks	Dense + Cascade (BGE→Q0.6B→Q4B)	0.554	0.722

Table 1: Effectiveness of all evaluated architectures on the ANEEL known-item benchmark, for document-level (top) and chunk-level (bottom) retrieval. MRR@10 and Recall@10 over 500 queries per granularity; under the known-item formulation, Recall@10 is a top-10 success rate. Best value per granularity and metric in **bold**. Retrieval-only rows return the first stage's top-10; "+ BGE" rows apply `bge-reranker-v2-m3` to the first stage's top-100. "2RR" denotes the two-stage Qwen reranking pipeline and "Cascade" the three-stage pipeline (Q0.6B/Q4B = `Qwen3-Reranker-0.6B/4B`).

reaches 0.755 MRR@10 and 0.934 Recall@10, outperforming both components; at the chunk level it reaches 0.432 MRR@10 and 0.698 Recall@10, with the same pattern. These results are consistent with complementary lexical and dense signals in this domain: BM25 captures exact regulatory terminology and formal references, while dense retrieval can contribute semantically related candidates even without surface-form overlap with the query.

5.3. RQ3: Single-stage versus multi-stage reranking

Applying a single BGE reranker improves ranking quality over first-stage retrieval. For documents, Dense+BGE improves MRR@10 from 0.656 to 0.792, BM25+BGE reaches 0.787, and Hybrid+BGE attains the highest observed MRR@10, 0.794. Differences among BGE configurations at the document level are small, showing that a strong reranker reduces part of the gap between first-stage strategies. Hybrid retrieval *without* reranking, however,

achieves the highest document-level Recall@10 (0.934, against 0.926 for Hybrid+BGE): reranking improves the ordering of top candidates but may push some relevant candidates out of the top 10. At the chunk level the benefit of combining hybrid retrieval with BGE is clearer: Hybrid+BGE achieves the best results on both metrics (0.566 MRR@10, 0.790 Recall@10), suggesting that the reranker benefits from a candidate set that combines lexical precision with semantic coverage. Chunk-level differences should also be read in light of the validation caveat of Section 3.3: 28% of the inspected chunk-level questions had some defect.

The Qwen-based multi-stage pipelines do not improve over the simpler BGE-based configurations. At the document level, the two-stage pipeline (0.644 MRR@10, 0.708 Recall@10) and the three-stage cascade (0.649, 0.712) fall below even the dense retrieval baseline. At the chunk level they approach the BGE configurations in MRR@10 (0.555 and 0.554) but reduce Recall@10 (0.736 and 0.722) and remain below Hybrid+BGE.

Table 2 adds the latency dimension for the pipelines evaluated with dense first-stage retrieval.

Granularity	**Architecture**	**MRR@10**	**R@10**	**s/query**
Docs	Dense (Qwen)	0.656	0.860	0.039
Docs	Dense (Qwen) + BGE	**0.792**	**0.922**	1.799
Docs	Dense + 2RR (Q0.6B→Q4B)	0.644	0.708	6.016
Docs	Dense + Cascade (BGE→Q0.6B→Q4B)	0.649	0.712	4.646
Chunks	Dense (Qwen)	0.364	0.610	0.049
Chunks	Dense (Qwen) + BGE	0.554	**0.760**	**1.187**
Chunks	Dense + 2RR (Q0.6B→Q4B)	**0.555**	0.736	1.875
Chunks	Dense + Cascade (BGE→Q0.6B→Q4B)	0.554	0.722	2.102

Table 2: Quality-latency comparison for pipelines using dense retrieval as the first stage, per granularity: MRR@10, Recall@10, and average seconds per query (all online stages, NVIDIA H100 80GB; see Section 4). Best value per granularity and column in **bold** (excluding the retrieval-only latency, which trivially has no reranking cost).

The single BGE reranker offers the best quality-latency trade-off among the measured configurations. For documents, Dense+BGE reaches 0.792 MRR@10 and 0.922 Recall@10 at 1.799 s/query, while the two-stage Qwen pipeline takes 6.016 s/query and performs below the dense baseline; the cascade is faster (4.646 s/query) but also below baseline on both metrics. For chunks, the deeper pipelines match Dense+BGE in MRR@10 while reduc-

ing Recall@10 and increasing latency (1.875 and 2.102 vs. 1.187 s/query). Figures 2 and 3 visualize this behavior.

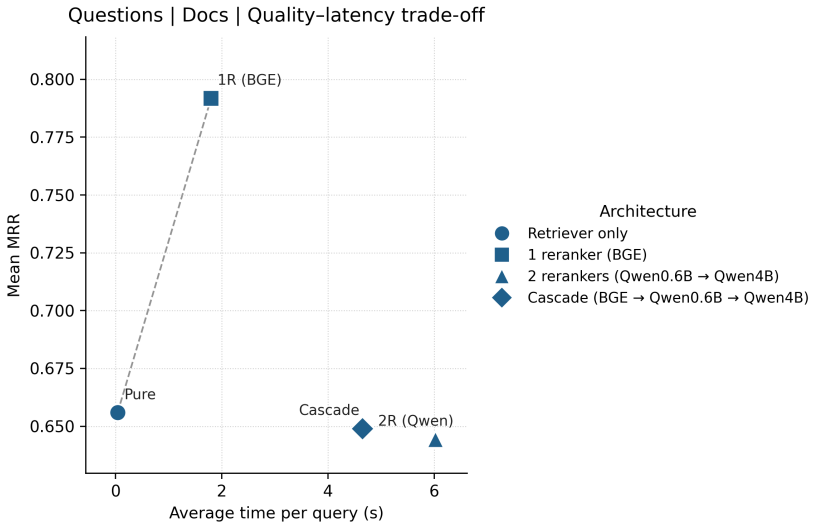


Figure 2: Document-level quality-latency trade-off for dense retrieval and the neural reranking pipelines of Table 2 (MRR@10 vs. average seconds per query; up and to the left is better). A single BGE reranker dominates the deeper pipelines, which cost more time and deliver lower effectiveness.

In summary: lexical retrieval is competitive and granularity-dependent (RQ1); hybrid fusion yields the strongest first-stage candidates at both granularities (RQ2); and a single BGE reranker over hybrid candidates achieves the best MRR@10 at both granularities and the best chunk-level Recall@10, while the evaluated deeper cascades do not justify their additional cost (RQ3). Document-level orderings rest on a single 500-query sample without variance estimates (Section 6.5).

6. Discussion

The results show that Portuguese regulatory search benefits from combining lexical, dense, and neural reranking signals — and that more complex neural pipelines are not automatically better. Each component contributes differently depending on retrieval granularity and pipeline stage.

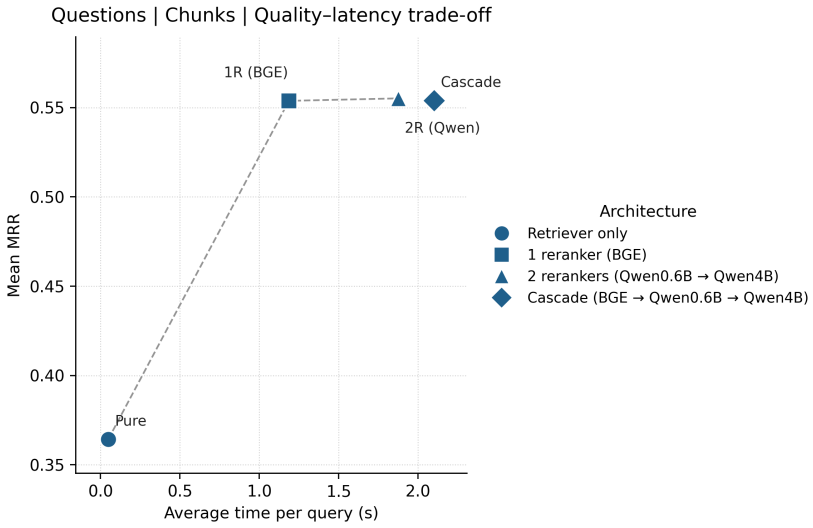


Figure 3: Chunk-level quality-latency trade-off for dense retrieval and the neural reranking pipelines of Table 2. Reranking improves MRR@10 over dense retrieval, but the deeper pipelines provide negligible MRR@10 gains over a single BGE reranker while reducing Recall@10 and increasing latency.

6.1. Why lexical evidence remains essential (RQ1, RQ2)

BM25's document-level advantage is plausibly linked to the structure of ANEEL resolutions, which contain stable terminology, resolution types, institutional references, dates, tariff categories, regulated entities, and formal expressions. In this domain, exact or near-exact lexical matches are not superficial signals; they often encode legally and operationally relevant information. At the same time, lexical retrieval alone is not sufficient on its own: users express information needs in natural language while target documents use formal legal-administrative wording, which is precisely the gap dense retrieval covers. The hybrid retriever's first-stage dominance at both granularities follows from this complementarity.

6.2. When reranking helps — and when depth hurts (RQ3)

A single BGE reranker improves top-rank ordering over every first-stage strategy, and is most effective when the candidate set combines lexical precision and semantic coverage (Hybrid+BGE). The document-level results also reveal a practical distinction between ranking quality and candidate recall: hybrid retrieval alone maximizes Recall@10 (0.934), while Hybrid+BGE

maximizes MRR@10 (0.794) at a small recall cost (0.926). If the deployment goal is that the correct document appears somewhere in the top results — for example, as the candidate pool of a RAG system — hybrid retrieval alone is already strong; if the goal is placing the correct result as high as possible in the ranking, adding BGE is preferable, provided its measured latency cost is acceptable.

The evaluated Qwen-based multi-stage pipelines underperformed the simpler configurations. Three factors plausibly explain this behavior. First, truncation: ANEEL resolutions often exceed the rerankers' context windows, so cross-encoders score truncated query-candidate pairs, and a larger model cannot help if the decisive evidence lies outside the observed context. Second, irreversible pruning: multi-stage pipelines assume early stages preserve relevant candidates; once an intermediate reranker demotes the correct candidate below its cutoff, later stages cannot recover it. A direct test of this hypothesis — measuring intermediate recall after each pruning step — remains future work. Third, zero-shot calibration: score distributions of off-the-shelf rerankers may transfer poorly to Brazilian regulatory text, where documents share templates, boilerplate, and legal vocabulary while relevance hinges on a specific article, tariff condition, deadline, or regulated entity. We emphasize the scope of this finding: the evaluated configurations differ in model family, size, ordering, cutoff schedule, and context length, so the evidence supports the narrower conclusion that *these* off-the-shelf Qwen-based cascades did not justify their cost in this benchmark — not that cascade depth is harmful in general.

6.3. Granularity matters (RQ1–RQ3)

Document-level and chunk-level retrieval are related but distinct tasks. At the document level, first-stage baselines are strong and the challenge is placing the correct resolution higher without losing top-10 recall. At the chunk level, first-stage retrieval is weaker and the task is to isolate one passage among many similar contextualized chunks — a reading consistent with the much larger reranking gains observed for chunks, though the experiment does not isolate the individual causes. The human validation supports this reading: all 25 sampled document-level questions were judged valid, while chunk-level questions concentrated the partial and bad cases, often answerable from the contextual prefix or the broader document rather than the specific chunk. Part of the chunk-level difficulty is therefore attributable to benchmark granularity rather than retrieval quality alone.

6.4. Practical implications

For institutional search over Brazilian regulatory documents, the evidence supports a hybrid-first design: keep BM25 (lexical evidence is highly informative), keep dense retrieval (semantic coverage), fuse with RRF, and apply a single lightweight reranker when top-rank ordering matters. This architecture is also operationally simpler than deeper cascades. More broadly, for long, formal, semi-structured collections — regulations, legal opinions, technical manuals, audit reports — retrieval evaluation should report effectiveness and latency jointly, include lexical and hybrid baselines, and treat architectural complexity as a hypothesis to test rather than a default improvement — recommendations that remain to be validated beyond the present benchmark.

6.5. Threats to validity

Four limitations bound the conclusions of this study; all four define the agenda for future work.

Single-target relevance.. The known-item formulation (Section 3.4) credits only the source item of each query. In a regulatory corpus, multiple resolutions or chunks may satisfy the same information need, so the reported metrics do not credit other potentially relevant items and may underestimate performance under multi-target judgments; Recall@10 must be read as a top-10 success rate. Extending the benchmark with multi-target relevance judgments — for example, via pooled LLM-assisted assessment with human audit [29, 31] — is the highest-priority extension.

Partial human validation.. The validation study covers 50 of 1,000 queries (5%), was performed by a single annotator — one of the authors — without independent adjudication, and 28% of the chunk-level sample was judged partial or bad. Metrics over the full query set therefore include an unquantified fraction of imperfect queries; the chunk-level results in particular should be interpreted with this margin in mind.

Sampling and variance.. The benchmark samples 500 documents of 18,364 and 500 chunks of 134,520, and no repeated runs or variance estimates are reported. Differences among the top document-level configurations are small (e.g., 0.794 vs. 0.792 MRR@10) and could plausibly be reordered under resampling or repeated runs. We therefore avoid claims of superiority among

closely ranked configurations and defer variance estimation (multiple samples, confidence intervals) to the extended study.

Corpus documentation.. The collection period and the crawling and preprocessing procedure of the ANEEL snapshot are not documented in this study, which limits exact reproduction of the corpus construction; the chunking parameters and benchmark construction steps of Section 3 are documented, and the released artifacts (see Data availability) mitigate but do not remove this gap.

Synthetic queries.. LLM-generated queries may not reflect real user behavior, and generation from the target passage risks lexical overlap that favors exact-match retrieval, despite prompt-level mitigation (Section 3.2). Complementing the benchmark with human-curated queries from institutional users remains future work.

7. Conclusion

We evaluated zero-shot retrieval and neural reranking pipelines for Portuguese regulatory information retrieval in the Brazilian electric power sector, using a documented LLM-generated known-item benchmark over ANEEL resolutions at two granularities. The three research questions can now be answered directly.

RQ1. Lexical and dense retrieval are granularity-dependent complements rather than competitors: on the evaluated 500-query sample, BM25 outperforms dense retrieval at the document level (0.677 vs. 0.656 MRR@10; 0.914 vs. 0.860 Recall@10 — under our known-item formulation, a top-10 success rate), while dense retrieval is marginally ahead in MRR@10 at the chunk level, with essentially equal Recall@10. This pattern is consistent with exact regulatory terminology acting as a first-class relevance signal in this domain.

RQ2. Hybrid sparse-dense retrieval with RRF provides the strongest first-stage candidate generation at both granularities (0.755 MRR@10 and 0.934 Recall@10 for documents; 0.432 and 0.698 for chunks), providing evidence that lexical and semantic signals were complementary in this evaluation.

RQ3. A single lightweight BGE reranker over hybrid candidates achieves the best observed effectiveness (0.794 MRR@10 for documents; 0.566 MRR@10 and 0.790 Recall@10 for chunks), with hybrid retrieval alone retaining the

best document-level Recall@10 (0.934). The evaluated Qwen-based multi-stage cascades, by contrast, increase latency up to 6.016 s/query without proportional gains, at the document level falling below the dense baseline. For the pipelines and setting evaluated here, deeper reranking was not justified as a default architectural choice.

The practical outcome is a benchmark-supported starting point for institutional retrieval over Brazilian regulatory documents: hybrid lexical-dense first-stage retrieval, followed by a single-stage BGE reranker when top-rank ordering matters. Three limitations qualify this guidance: only 50 of the 1,000 synthetic queries were manually inspected; 28% of the inspected chunk-level questions had some defect, so chunk-level conclusions in particular should be read with caution; and document-level orderings rest on a single 500-query sample without variance estimates, so small differences among top configurations should not be over-read. Future work follows the agenda of Section 6.5: multi-target relevance judgments with LLM-assisted assessment under human audit, broader human validation of the synthetic queries, variance estimation across samples and runs, intermediate-recall analysis of cascade pruning, and extensions to additional regulatory agencies, document families, and real user queries, as well as long-context reranking and passage-aware document scoring.

Data availability

The benchmark queries, target identifiers, generation code (including the query-generation prompt), and evaluation code will be made available upon acceptance.

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Appendix A. Query generation prompt: characterization

The queries were generated with a single prompt template, instantiated once per sampled item with the item's text. The prompt instructs the model to produce exactly one self-contained question that (i) sounds like a query a real user would issue; (ii) can be answered from the provided text alone; (iii) avoids meta-references such as "text", "excerpt", "document", "above", or "below"; (iv) does not copy phrases from the source, use internal identifiers, or depend on seeing the passage; and (v) follows one of two requested types, generated in balanced proportion: *general* questions about the broader topic, objective, or regulatory matter of the target, and *specific* questions about concrete details such as deadlines, conditions, obligations, responsible entities, consequences, or exceptions. The verbatim prompt string is part of the generation code covered by the data availability statement.

For illustration, two queries from the chunk-level benchmark are shown below with their target identifiers. The first asks about the broad purpose of the target passage; the second asks for concrete values.

> **Example 1** (`rea202110739ti_c000002`): *"Qual o objetivo da desapropriação de terras em Nova Venécia em favor da EDP Espírito Santo?"*
>
> **Example 2** (`reh2009858_c000004`): *"Quais eram as tarifas de energia e demanda da Iguaçu Distribuidora para o subgrupo A4 vigentes entre 2009 e 2010?"*

References

[1] M. van Opijnen, C. Santos, On the concept of relevance in legal information retrieval, Artificial Intelligence and Law 25 (1) (2017) 65–87.

[2] C. Sansone, G. Sperlí, Legal information retrieval systems: State-of-the-art and open issues, Information Systems 106 (2022) 101967.

[3] S. Robertson, H. Zaragoza, The probabilistic relevance framework: BM25 and beyond, Foundations and Trends in Information Retrieval 3 (4) (2009) 333–389.

[4] V. Karpukhin, B. Oğuz, S. Min, P. Lewis, L. Wu, S. Edunov, D. Chen, W.-t. Yih, Dense passage retrieval for open-domain question answering, in: Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP), 2020, pp. 6769–6781.

[5] J. Lin, R. Nogueira, A. Yates, Pretrained Transformers for Text Ranking: BERT and Beyond, Morgan & Claypool, 2021.

[6] G. V. Cormack, C. L. A. Clarke, S. Buettcher, Reciprocal rank fusion outperforms Condorcet and individual rank learning methods, in: Proceedings of the 32nd International ACM SIGIR Conference on Research and Development in Information Retrieval, 2009, pp. 758–759.

[7] S. Bruch, S. Gai, A. Ingber, An analysis of fusion functions for hybrid retrieval, ACM Transactions on Information Systems 42 (1) (2023).

[8] E. de Oliveira, M. Bueno, R. Nogueira, R. Lotufo, J. Pereira, Quati: A Brazilian Portuguese information retrieval dataset from native speakers, in: Proceedings of the Symposium in Information and Human Language Technology (STIL 2024), 2024, pp. 185–195.
URL https://aclanthology.org/2024.stil-1.19

[9] L. C. Fernandes, L. d. S. Ribeiro, M. V. B. de Castro, L. A. da Silva Pacheco, E. F. de Oliveira Sandes, JurisTCU: A Brazilian Portuguese information retrieval dataset with query relevance judgments, Language Resources and Evaluation 60 (1) (2026) 23.

[10] J. Pereira, L. Fernandes, E. de Brito, R. Lotufo, L. Bonifacio, JUÁ: A benchmark for information retrieval in Brazilian legal text collections, arXiv:2604.06098 (2026).

[11] R. Nogueira, K. Cho, Passage re-ranking with BERT, arXiv:1901.04085 (2019).

[12] G. Capannini, C. Lucchese, F. M. Nardini, S. Orlando, R. Perego, N. Tonellotto, Quality versus efficiency in document scoring with learning-to-rank models, Information Processing & Management 52 (5) (2016) 902–918.

[13] R. Nogueira, W. Yang, K. Cho, J. Lin, Multi-stage document ranking with BERT, arXiv:1910.14424 (2019).

[14] Y. Sasazawa, K. Yokote, O. Imaichi, Y. Sogawa, Text retrieval with multi-stage re-ranking models, arXiv:2311.07994 (2023).

[15] K. Zheng, H. Zhao, R. Huang, B. Zhang, N. Mou, Y. Niu, Y. Song, H. Wang, K. Gai, Full stage learning to rank: A unified framework for multi-stage systems, in: Proceedings of the ACM Web Conference 2024, 2024.

[16] J. S. Rayo Mosquera, C. R. De La Rosa Peredo, M. Garrido Cordoba, A hybrid approach to information retrieval and answer generation for regulatory texts, in: Proceedings of the 1st Regulatory NLP Workshop (RegNLP 2025), COLING 2025, 2025, pp. 31–35.
URL https://aclanthology.org/2025.regnlp-1.5

[17] I. Chalkidis, M. Fergadiotis, P. Malakasiotis, N. Aletras, I. Androutsopoulos, LEGAL-BERT: The muppets straight out of law school, in: Findings of the Association for Computational Linguistics: EMNLP 2020, 2020, pp. 2898–2910.

[18] I. Chalkidis, A. Jana, D. Hartung, M. Bommarito, I. Androutsopoulos, D. Katz, N. Aletras, LexGLUE: A benchmark dataset for legal language understanding in English, in: Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), 2022, pp. 4310–4330.

[19] R. Malaquias Junior, R. Pires, R. Romero, R. Nogueira, Juru: Legal Brazilian large language model from reputable sources, arXiv:2403.18140 (2024).

[20] R. Nogueira, Z. Jiang, R. Pradeep, J. Lin, Document ranking with a pretrained sequence-to-sequence model, in: Findings of the Association for Computational Linguistics: EMNLP 2020, 2020, pp. 708–718.

[21] J. Guo, Y. Fan, L. Pang, L. Yang, Q. Ai, H. Zamani, C. Wu, W. B. Croft, X. Cheng, A deep look into neural ranking models for information retrieval, Information Processing & Management 57 (6) (2020) 102067.

[22] S. Althammer, A. Askari, S. Verberne, A. Hanbury, DoSSIER@COLIEE 2021: Leveraging dense retrieval and summarization-based re-ranking for case law retrieval, in: Proceedings of the Eighth International Competition on Legal Information Extraction/Entailment (COLIEE 2021), 2021, pp. 8–14.

[23] M.-Y. Kim, J. Rabelo, R. Goebel, M. Yoshioka, Y. Kano, K. Satoh, COLIEE 2022 summary: Methods for legal document retrieval and entailment, in: JSAI International Symposium on Artificial Intelligence, Springer, 2022, pp. 51–67.

[24] J. Zhu, X. Luo, J. Wu, A BERT-based two-stage ranking method for legal case retrieval, in: Challenge of AI in Law (CAIL 2021), Springer, 2022, pp. 534–546.

[25] A. M. A. Novais, A. P. V. L. B. Moreira, M. C. X. d. Almeida, J. P. C. Presa, F. M. Federson, S. S. T. d. Oliveira, Optimizing efficiency in multi-stage semantic re-ranking architectures, in: Proceedings of the 17th International Conference on Computational Processing of Portuguese (PROPOR 2026) - Vol. 1, 2026, pp. 562–571.
URL https://aclanthology.org/2026.propor-1.55/

[26] L. Bonifacio, V. Jeronymo, H. Abonizio, I. Campiotti, M. Fadaee, R. Lotufo, R. Nogueira, mMARCO: A multilingual version of the MS MARCO passage ranking dataset, arXiv:2108.13897 (2021).

[27] L. Bonifacio, H. Abonizio, M. Fadaee, R. Nogueira, InPars: Unsupervised dataset generation for information retrieval, in: Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval, 2022, pp. 2387–2392.

[28] Z. Dai, V. Y. Zhao, J. Ma, Y. Luan, J. Ni, J. Lu, A. Bakalov, K. Guu, K. Hall, M.-W. Chang, Promptagator: Few-shot dense retrieval from 8 examples, in: Proceedings of the Eleventh International Conference on Learning Representations (ICLR), 2023.

[29] P. Thomas, S. Spielman, N. Craswell, B. Mitra, Large language models can accurately predict searcher preferences, in: Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval, 2024, pp. 1930–1940.

[30] G. Faggioli, L. Dietz, C. L. A. Clarke, G. Demartini, M. Hagen, C. Hauff, N. Kando, E. Kanoulas, M. Potthast, B. Stein, H. Wachsmuth, Perspectives on large language models for relevance judgment, in: Proceedings of the 2023 ACM SIGIR International Conference on Theory of Information Retrieval (ICTIR), 2023, pp. 39–50.

[31] S. Upadhyay, R. Pradeep, N. Thakur, D. Campos, N. Craswell, I. Soboroff, H. T. Dang, J. Lin, A large-scale study of relevance assessments with large language models using UMBRELA, in: Proceedings of the 2025 International ACM SIGIR Conference on Innovative Concepts and Theories in Information Retrieval (ICTIR), 2025. doi:10.1145/3731120.3744605.

[32] N. Thakur, N. Reimers, A. Rücklé, A. Srivastava, I. Gurevych, BEIR: A heterogeneous benchmark for zero-shot evaluation of information retrieval models, in: Proceedings of the Thirty-fifth Conference on Neural Information Processing Systems, Datasets and Benchmarks Track, 2021.

[33] Anthropic, Introducing contextual retrieval, `https://www.anthropic.com/news/contextual-retrieval` (2024).

[34] Gemini Team, Google, Gemini: A family of highly capable multimodal models, arXiv:2312.11805 (2023).

[35] Y. Zhang, M. Li, D. Long, X. Zhang, H. Lin, B. Yang, P. Xie, A. Yang, D. Liu, J. Lin, F. Huang, J. Zhou, Qwen3 embedding: Advancing text embedding and reranking through foundation models, arXiv:2506.05176 (2025).

[36] J. Johnson, M. Douze, H. Jégou, Billion-scale similarity search with GPUs, IEEE Transactions on Big Data 7 (3) (2021) 535–547.

[37] J. Chen, S. Xiao, P. Zhang, K. Luo, D. Lian, Z. Liu, M3-embedding: Multi-linguality, multi-functionality, multi-granularity text embeddings through self-knowledge distillation, in: Findings of the Association for Computational Linguistics: ACL 2024, 2024, pp. 2318–2335.
URL `https://aclanthology.org/2024.findings-acl.137/`